



TECHATHON 1.0

COMBAT DRONE

IUT
ROBOTICS
SOCIETY

Jilapi

SITUATION ANALYSIS

MARKET INSIGHT FOR COMBAT DRONE:

- ❖ **Increased Demand:** Growing global conflicts and the demand for advanced surveillance are boosting the need for military drones.
- ❖ **Technological Advancements:** Rapid drone tech advancements, like AI autonomy, are boosting their combat effectiveness.
- ❖ **Cost Efficiency:** Drones provide cost-effective alternatives to manned aircraft for intelligence, target acquisition, and strikes.
- ❖ **Operational Flexibility:** Drones offer military forces flexibility for operation in varied environments, reducing personnel risk.
- ❖ **Global Market Growth:** The global military drone market is poised for continued growth, fueled by ongoing conflicts, territorial disputes and global defense modernization efforts

OBJECTIVES

- Surveillance & Reconnaissance.
- Target Acquisition.
- Precision Strikes.
- Force Enhancement.
- Marketing & Monetization.

STATUS:

- ❖ Provide B2B and B2G service



PROBLEM IDENTIFICATION



A

Durability and Protection

Combat drones require reinforced structures and protective armor.

B

Weapon Integration

Mounting weapons onto drones while ensuring stability, accuracy, and safety.

CORE ISSUE

Enhancing navigation systems to support combat missions and increasing payload capacity

Power Supply and Management

Combat drones need robust and reliable power sources.

C

Communication and control

Secure and reliable communication links between the drone and its operator are mandatory.

D

TARGET GROUP ANALYSIS

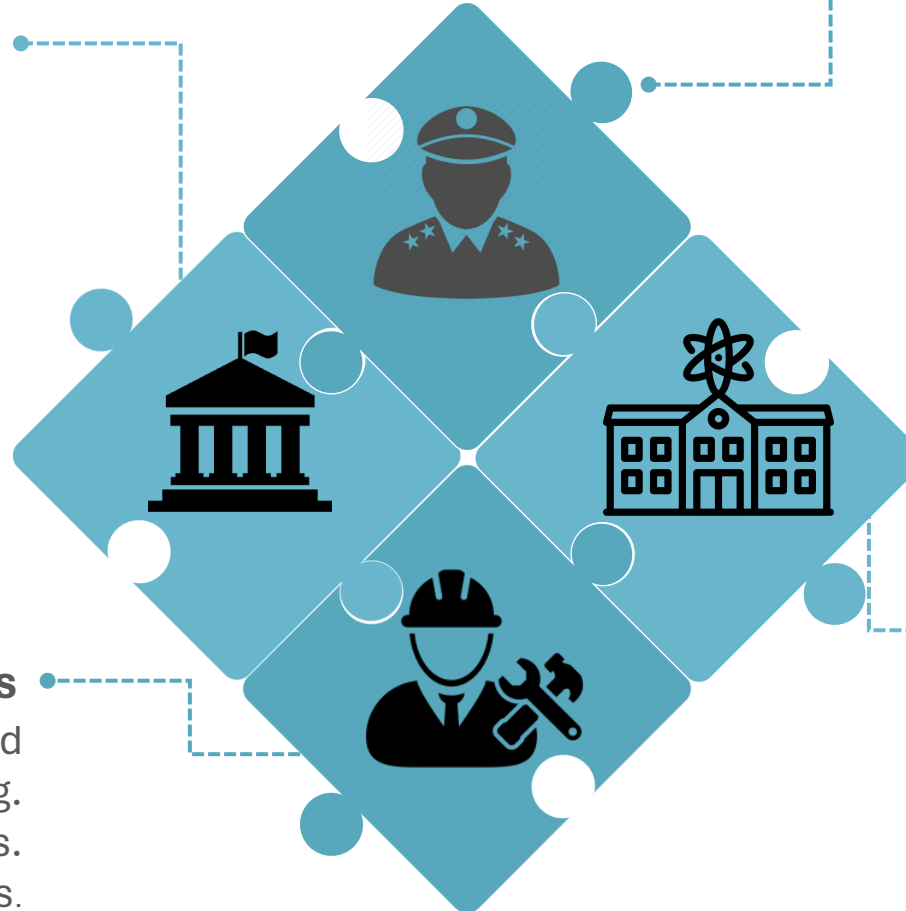


Government Agencies

- Oversee the procurement process for combat drone programs.
- Enforce regulations and provide oversight.
- Ensure combat drone programs comply with established national security

Defence Contractors

- Specialize in designing and manufacturing.
- Provide maintenance services.
- Cater exclusively to military clients.



Military Forces

- Primary users of combat drones for surveillance, reconnaissance and strike mission
- Enhance operational capabilities
- Reduce risk to personnel.

Research Institutions

- Focus on advancing drone technology.
- Aim to improve autonomy, payload and counter drone system.
- Often collaborate with industry partners and govt. agencies.

BIG IDEA



Since we lack the resources to produce combat drones in Bangladesh, we're exploring alternative options. We plan to reach out to "**Baykar**" a renowned drone manufacturer in **Turkey**. Our goal is to **establish a partnership** with them to modify their drones into combat drones according to our needs.

We aim to negotiate a deal with "**Baykar**", potentially **hiring their researchers** on a temporary basis to work with us and facilitate the conversion process.

Unveil the
Art of
Defense
with Arial
Mastery

Tagline

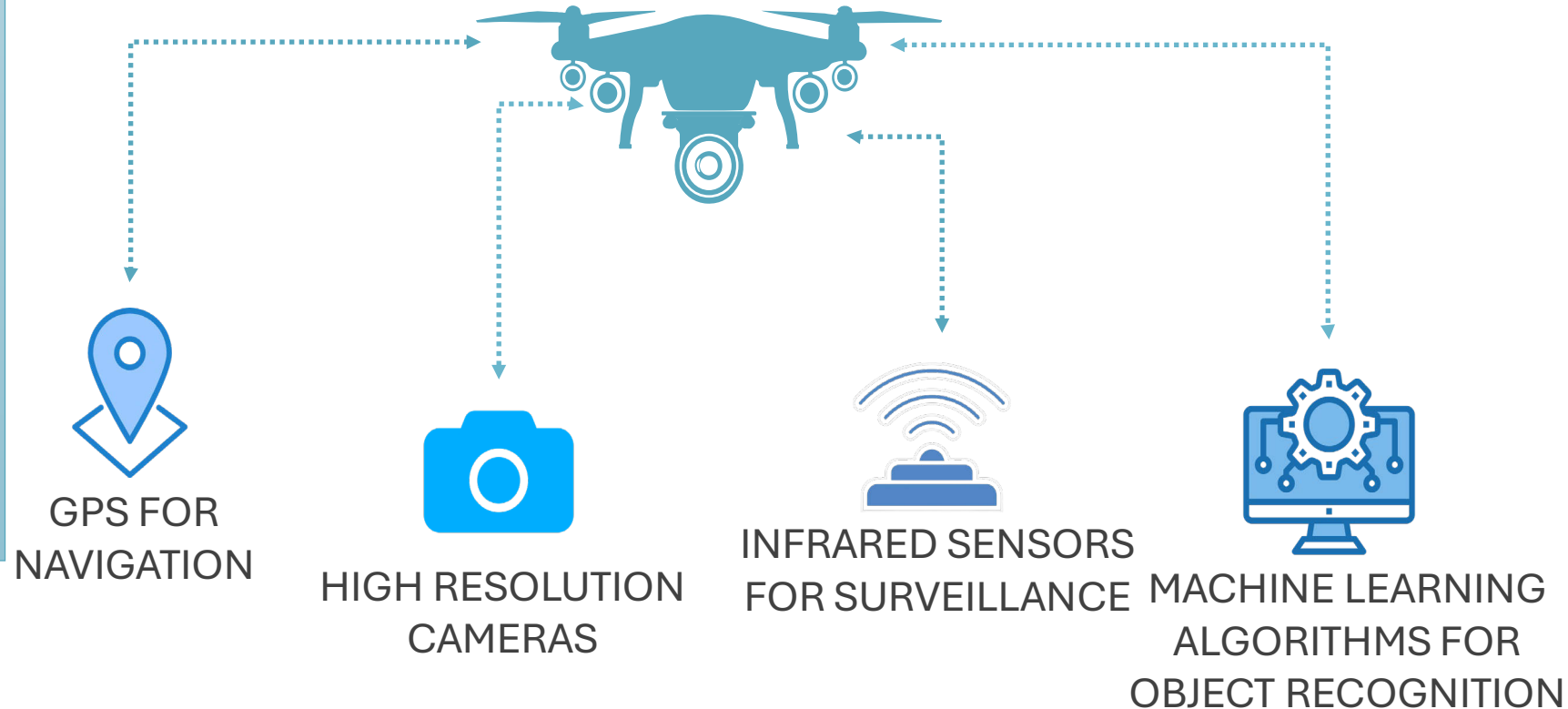


ACTION PLAN



To convert a regular drone into a "Bayraktar TB2 drone" for military purposes, we'll need specific instruments:

The "Bayraktar TB2" is a medium-altitude long-endurance (MALE) unmanned combat aerial vehicle (UCAV) capable of autonomous or remotely controlled flight operations. So we want to change a normal drone to "bayraktar TB2 drone" for war purpose.



ACTION PLAN



★ Battlefield damage assessment using drones involves several steps and technologies:

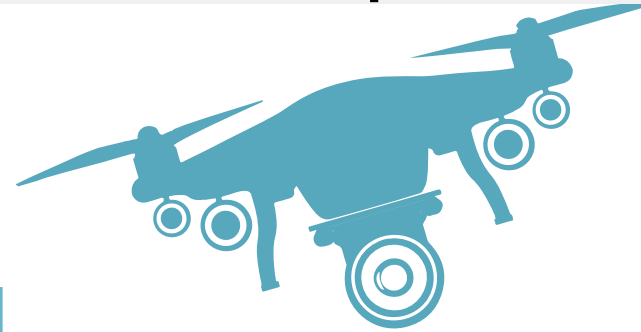


Pre-Mission Planning

Define the area of interest and establish objectives for the assessment

Aerial Surveying

Drones equipped with high-resolution cameras, LIDAR, or other sensors conduct aerial surveys of the battlefield



Data Collection and Analysis

We will analyze the collected data to identify and quantify the extent of damage

Classification Of Damage

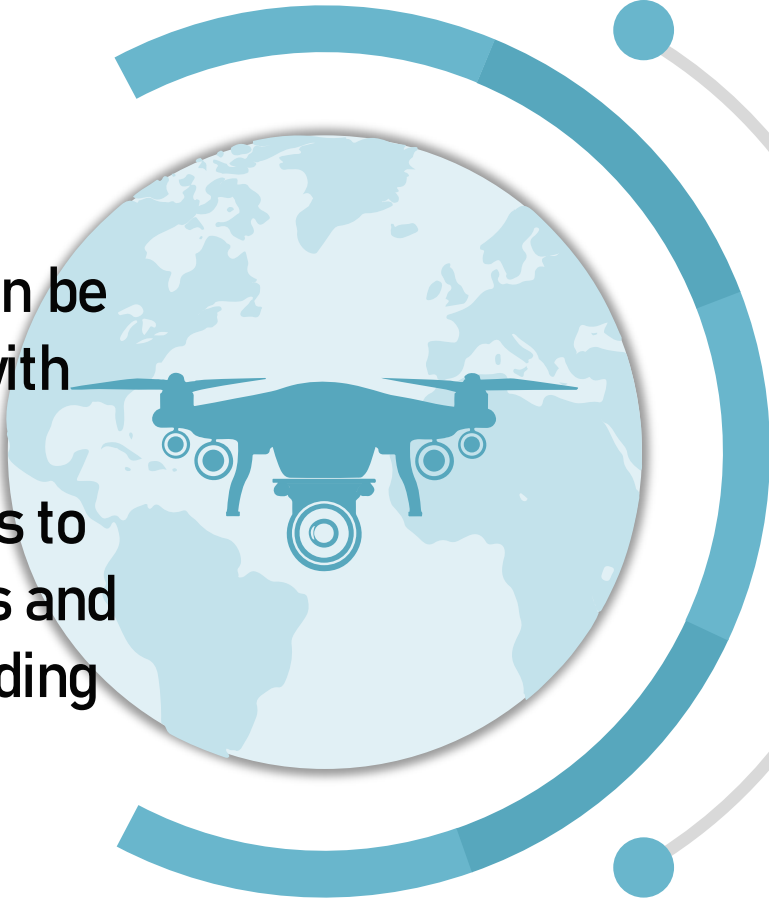
We will turn the damage into categories such as destroyed, severely damaged, moderately damaged or unaffected



ACTION PLAN



★ Drones can be equipped with various technologies to detect bombs and mines, including



Metal Detectors

Drones can carry metal detectors to identify metallic components of bombs and mines

Thermal Imaging

Thermal cameras on drones can detect differences in temperature

Chemical Sensors

Drones can carry chemical sensors capable of detecting explosive materials or gases

Laser-based System

LIDAR (Light Detection and Ranging) system can be used to create detailed 3D maps of the terrain

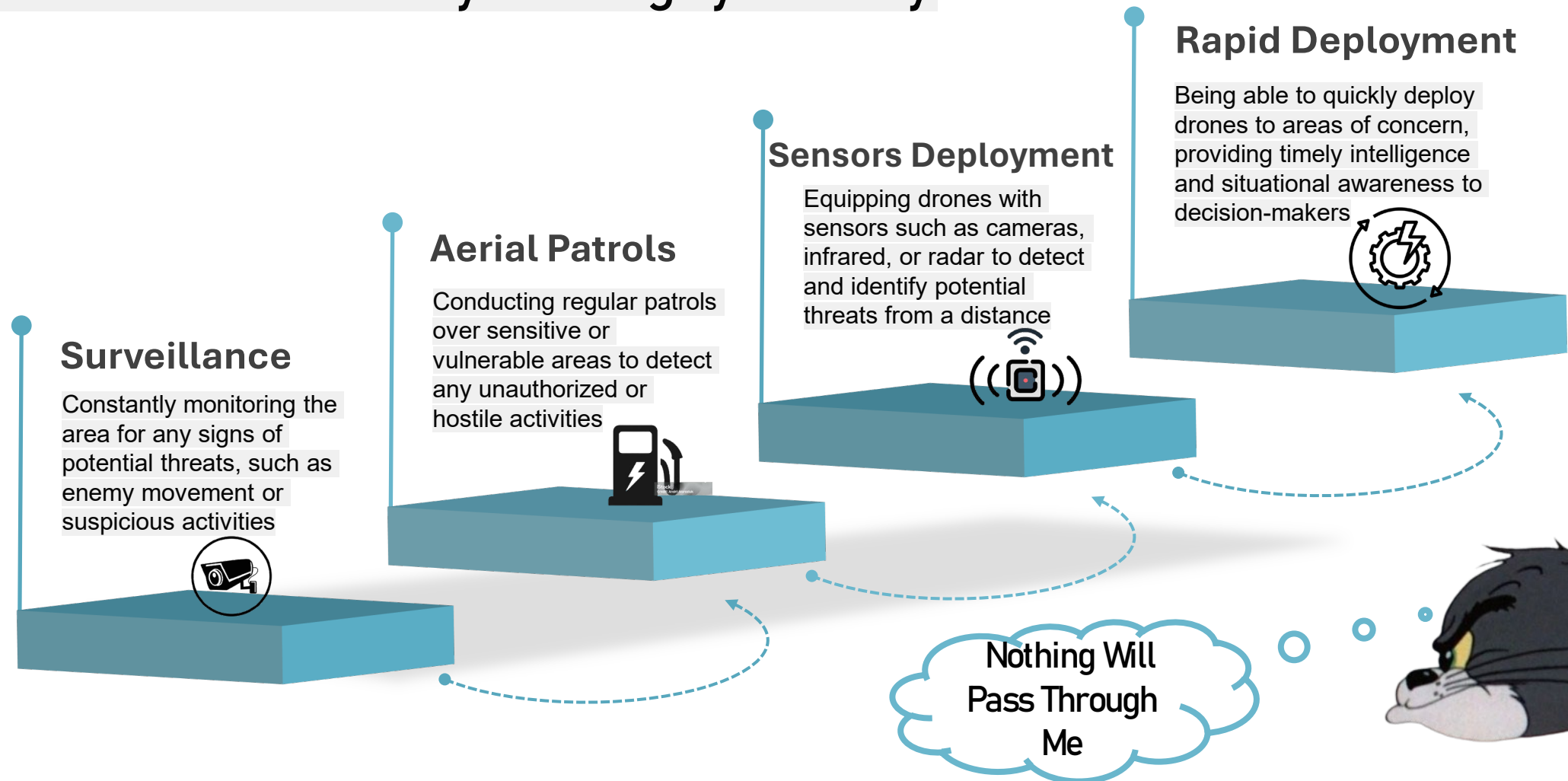
Multi-Spectral imaging

This technology combines various imaging techniques, such as infrared and ultraviolet

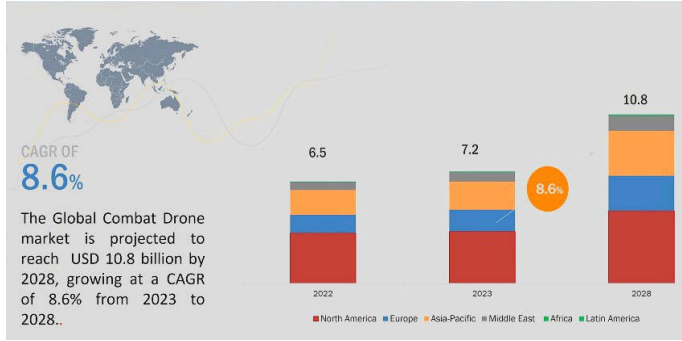
ACTION PLAN



Drones can serve as early warning systems by:



MARKETING PLAN



AUGMENTED REALITY (AR) EXPERIENCES

- Create AR apps for users to visualize combat drones.
- Enable users to interact with virtual drone models, exploring features in detail for immersive experiences.

INFLUENCER MARKETING CAMPAIGNS

- Partner with military influencers, retired commanders, and veterans on social media
- Utilize their credibility and authority to build trust



INTERACTIVE DEMONSTRATION

- Organize live demonstrations or virtual reality experiences
- Allow potential customers to interact with the drones

WEB BASED MARKETING

Content Marketing:

Share regular content on combat drones

Social Media Marketing:

Leverage LinkedIn, Twitter, and YouTube to showcase combat drone products



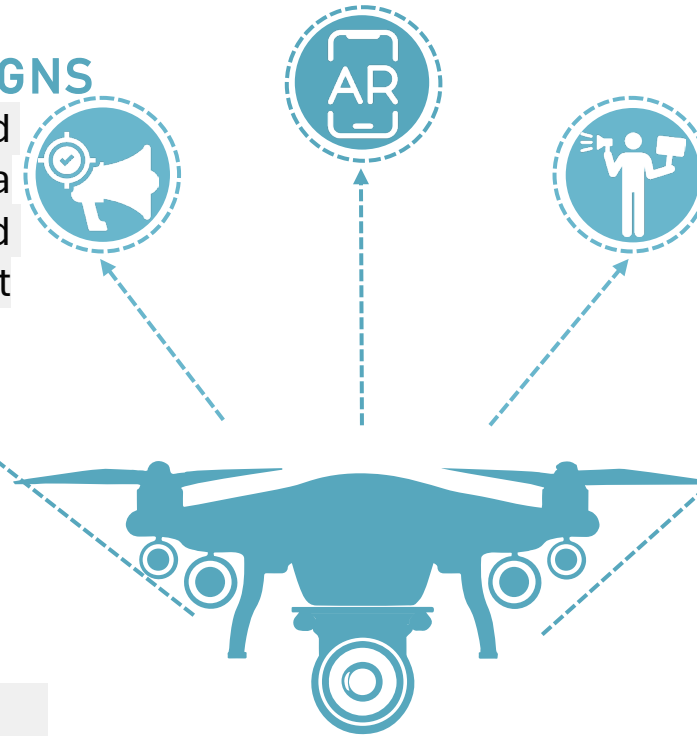
E-COMMERCE EXPANSION

Website Development:

Website Development: Develop an e-commerce site featuring combat drone products

Marketing and Promotion:

Promote the e-commerce store with targeted campaigns



TECHNICAL SUSTAINIBILITY



MAINTENANCE AND SUPPORT



***Maintenance system:**

Implement regular inspections and scheduled maintenance routines

***Prompt Troubleshooting:**

Offer swift troubleshooting services to minimize downtime

***Lifecycle Optimization:**

Establish a comprehensive support framework to ensure peak performance

SOFTWARE UPDATES



***Software Enhancement:**

Create a reliable mechanism for deploying patches, enhancements

***Addressing Vulnerabilities:**

Ensure timely updates to address vulnerabilities

***Operational Flexibility:**

Enhance functionality and adaptability of combat drones

TESTING AND VALIDATION



***Rigorous Testing:**

Conduct thorough testing procedures

***Verify Reliability:**

Ensure reliability, durability, and safety

***Performance**

Evaluation:

Evaluate performance across various mission scenarios

ENVIRONMENT IMPACT



***Environmental Considerations:**

Incorporate energy efficiency and emissions reduction

***Sustainable Materials:**

Source materials sustainably to reduce ecological footprints.

***Regulatory Compliance:**

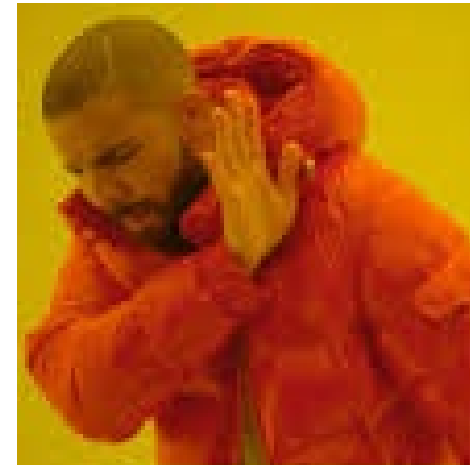
Ensure compliance with environmental regulations

FINANCIAL FEASIBILITY



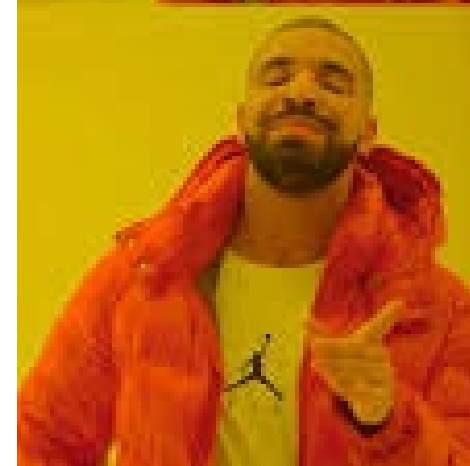
MARKETING COST AND DRONE EXPENDATURE

Material/Events	Cost/Unit (BDT)	Unit	Month	Total Cost
Marketing and Promotion	2 lac	1		2 lac
Web-based Marketing	1 Lac	1	3	3 Lac
E-commerce Expansion	5 Lac	1	2	10 Lac
Influencer Marketing	10,000	25	3	7,50,000
Drone Cost	8 core	5		40 core
			Grand Total=	40,22,50,000/=



In Turkey
(purchase cost)

Cost for production of 1
drone = 10 core (BDT)



In Bangladesh
(purchase cost)

Cost for production of 1
drone = 8 core (BDT)